LLM-Based with hybrid ML approach Classification of Symmetric Encryption Algorithms from Ciphertext

CipherScope

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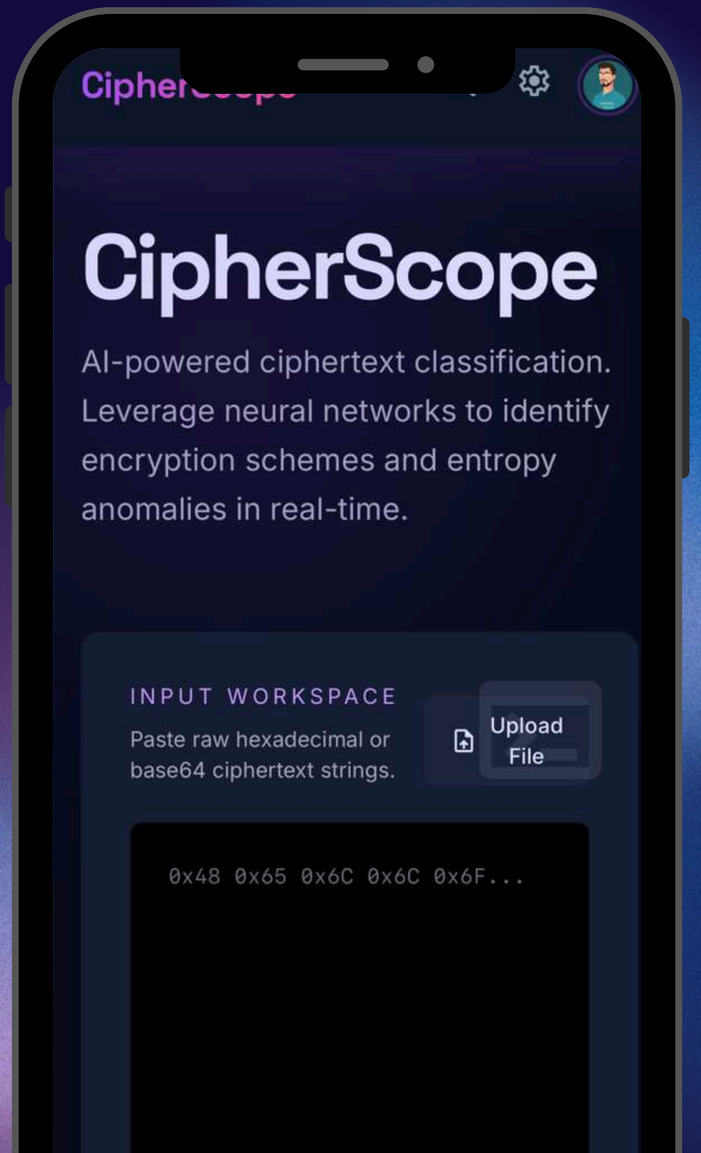
https://cipherscope.onrender.com/

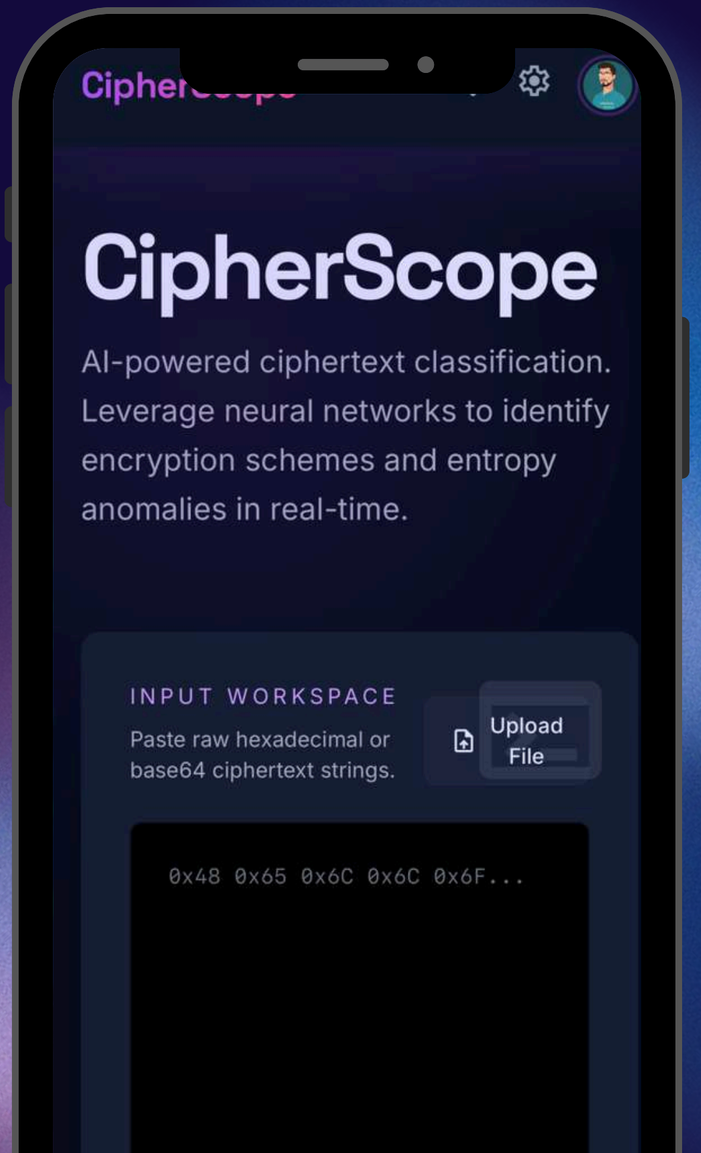
Agenda

- Problem Statement
- Introduction
- System Overview
- Results
- Demo : Cipherscope
- Limitations
- Future work



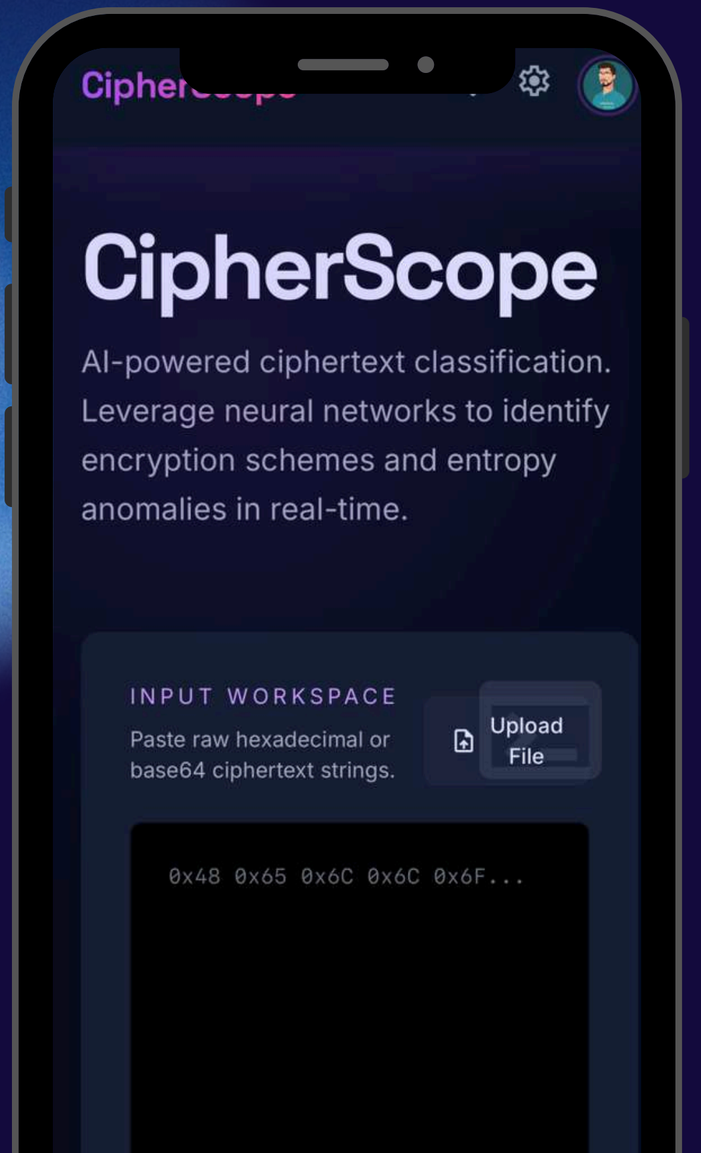
Problem Statement

- Can we identify the encryption algorithm using only ciphertext?
- No access to:
  - Encryption key
  - Plaintext
- Goal: Build an AI model to classify encryption algorithms



Introduction

- CipherScope is an LLM-based with hybrid ML approach platform designed for cryptographic analysis
- It focuses on identifying ciphertext encryption algorithms
- Specifically targets symmetric encryption algorithms
- The system analyzes ciphertext patterns using a trained model
- It predicts which algorithm was used without needing the encryption key or original data



System Overview

Our code takes ciphertext as input, extracts statistical features from it, and uses a trained machine learning model to predict the most likely encryption algorithm used.

Ciphertext input

HEX to Byte Conversion

Feature Extraction

Feature Vector Creation

Model Training

Output: Predicted Algorithm

Ciphertext input	8f434346648f6b96df89dda901c5176b
HEX to Byte Conversion	[143, 67, 67, 70, 100, ...]
Feature Extraction	Entropy: 3.625 Byte Variance: 0.00032 Chi-square: 21.0 Patterns: Transition relationships between bytes
Feature Vector Creation	[3.625, 0.00032, 21.0, 16, 2.75, 0.0625, 1.0, ...]
Model Training (XGBoost)	
Output: Predicted Algorithm	Prediction : AES

Example

Result

```
Dataset size: 25000
Extracting features...
Training model...

Classification Report:

              precision    recall  f1-score   support

         AES       1.00      1.00      1.00      1024
      AFFINE       1.00      1.00      1.00       950
  DES_FAMILY       1.00      1.00      1.00      2014
         RC4       1.00      1.00      1.00      1012

    accuracy                           1.00      5000
   macro avg       1.00      1.00      1.00      5000
weighted avg       1.00      1.00      1.00      5000


Confusion Matrix:

[[1024    0    0    0]
 [   0  950    0    0]
 [   0    0 2014    0]
 [   0    0    0 1012]]

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LLM-Based Classification of Symmetric Encryption Algorithms from Ciphertext

Demo : CipherScope

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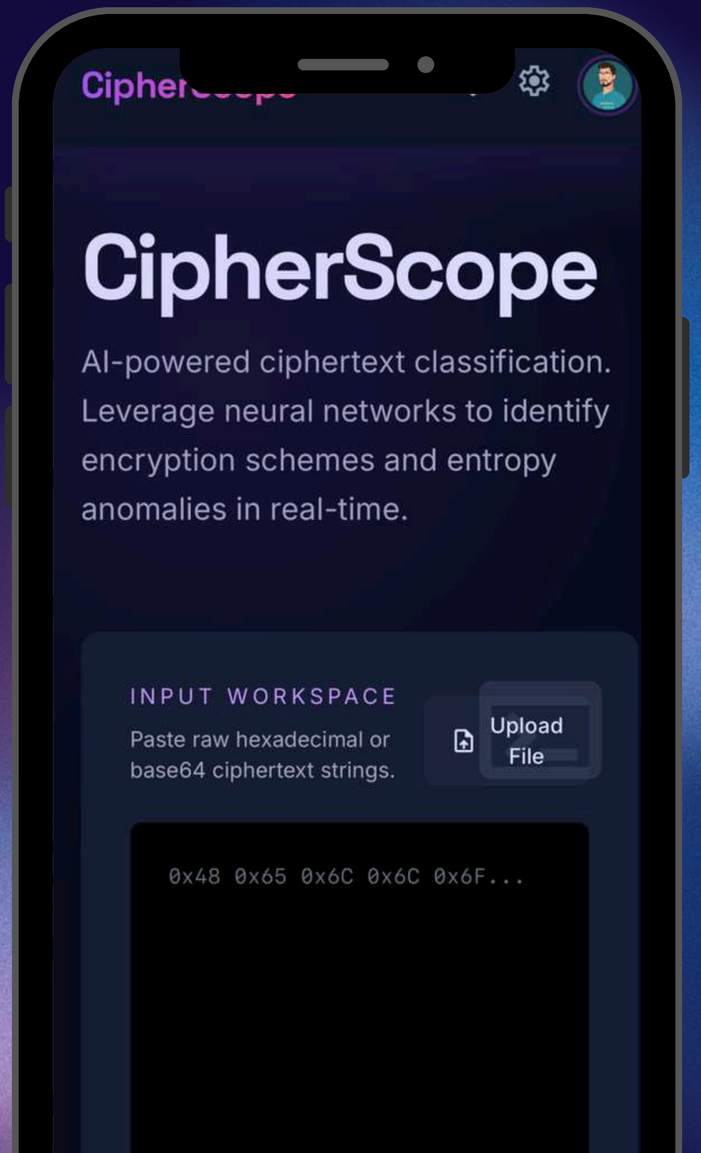
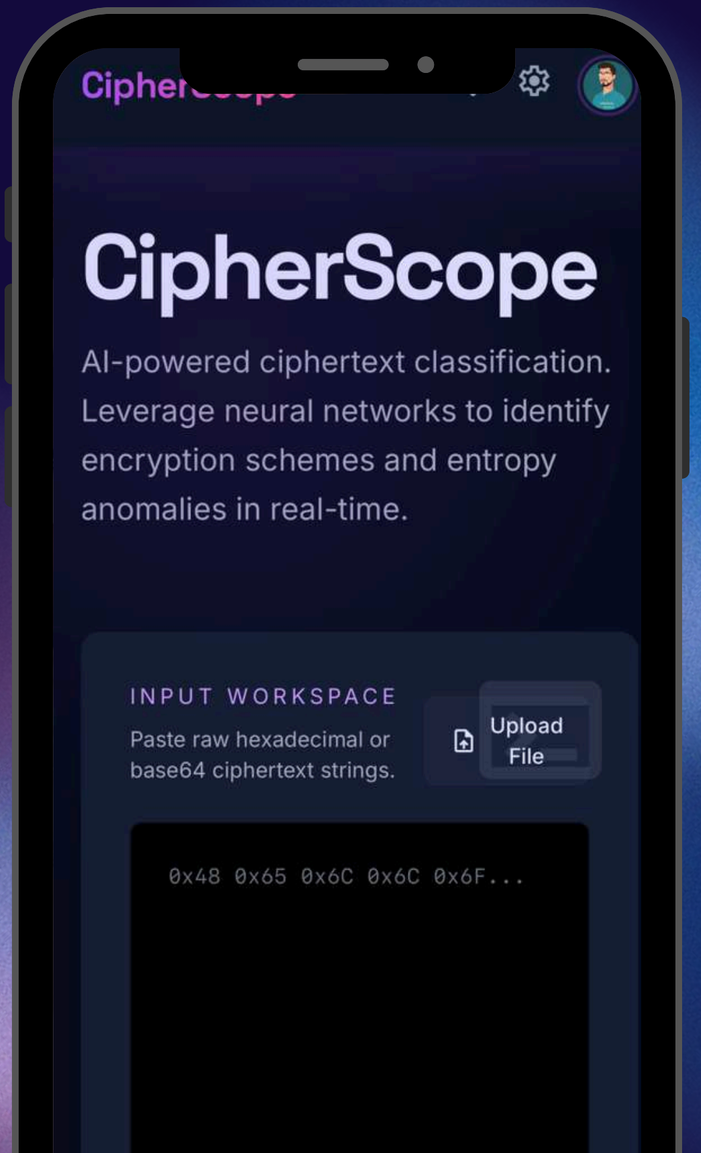
Limitations

- Ciphertext is designed to be highly random, making classification difficult
- AES and DES/3DES can have very similar statistical properties
- Model depends heavily on quality and size of dataset
- Performance may drop on unseen encryption modes or real-world data
- Cannot identify encryption key or mode (ECB, CBC, etc.)
- Risk of overfitting on training data patterns



Future Work

- Generate a larger and more diverse dataset to improve model learning
- Train the model on varied ciphertext samples for better generalization
- Build a system capable of detecting algorithms from any given ciphertext
- Improve model accuracy for real-world encrypted data
- Enhance robustness to handle different encryption styles and inputs



Thank you!

Questions?

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